

Practical 6 - Competing risks 1

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

Datasets required: `aaatrial.dta`

Introduction

In this practical we will use a dataset from a randomised trial of screening for abdominal aortic aneurysms (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The variables used in this practical are described below.

Variable	Description
id	Unique identifier for each participant
group	Randomisation group: 0=Control, 1=Invited
dateran	Date of randomisation
aaadeath	AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death
allddeath	All-cause mortality: 0=Censored, 1=Death (any cause)
deathtype	0=Censored; 1=non-AAA death; 2=AAA-death
timeout	Date of study exit

Aims

The aims of this session are:

- Estimate cumulative incidence functions using non-parametric methods (the Aalen-Johansen estimator), and plot and interpret these.
- Compare cumulative incidence functions between groups.
- Understand why the Kaplan-Meier estimator is not in general suitable for estimating cumulative incidences when there are competing events.

Question 1

- Load the data.
- How many people died from any cause, and how many had an AAA-death and a non-AAA-death?

Question 2

We will begin by focusing on death from any cause using the `allddeath` variable.

- Use the following `stset` code:

```
stset timeout, fail(allddeath) origin(dateran) id(id) scale(365.25)
```

- Plot the Kaplan-Meier survival curves for all-cause mortality.
- What is the risk of death from any cause up to 5 years?

Question 3

We will now move on to considering AAA-deaths and non-AAA-deaths as competing events. In Stata re-stset your data using the `deathtype` failure variable as shown below, indicating AAA-death as failure. Study the output from `stset` and how it differs from that in question 2.

Take a look at `_t` and `_d` for the first three individuals in the data. What information is given for (i) people who have an AAA-death, (ii) people who have a non-AAA-death, (iii) people who are censored?

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)
```

Question 4

- Estimate the cumulative incidences for AAA-deaths and non-AAA-deaths using the following code

```
**You may need to install the stcompet function like this  
ssc install stcompet  
  
** Generate the cumulative incidences for each event type  
stcompet cuminc=ci, compet1(1)  
  
gen cuminc_AAA = cuminc if deathtype==2  
gen cuminc_nonAAA = cuminc if deathtype==1
```

- The resulting cumulative incidence curves can be plotted using

```
twoway (line cuminc_AAA _t if deathtype==2, connect(step) sort lcolor(black)) ///  
(line cuminc_nonAAA _t if deathtype==1, connect(step) sort lcolor(black) lpattern(dash)) ///  
, legend(lab(1 "Controls") lab(2 "Invited")) title("Cumulative incidence of AAA deaths")
```

- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years?

Question 5

- Returning to considering death from any cause (**alldeath**), obtain Kaplan-Meier plots by treatment group.
- What is the risk of death from any cause up to 5 years in the two groups?
- Use a log-rank test to test the null hypothesis that there is no difference between the survival curves in the two groups.

Question 6

We now return to considering the competing events of AAA-death and non-AAA death.

- Estimate the cumulative incidence function for AAA-deaths and non-AAA-deaths in each treatment group by modifying the code used in question 4 - obtain the cumulative incidences using **stcompet** separately by group.
- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years in the two groups?
- A test of the null hypothesis of no difference in the cumulative incidences for a given cause of death by group can be performed as follows:

```
stpepemori group, compet(1)
```

Question 7

- Use the Kaplan-Meier estimator to estimate the cumulative incidence of AAA-death using

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)  
sts graph, failure ylabel(minmax)
```

- Use the code in the do file to plot the cumulative incidence based on the Kaplan-Meier estimate and compare this with what you obtained in Question 4 based on the Aalen-Johansen estimator. Comment on why the results are different.